

1. Graph $f(x) = -2(x - 3)^2 + 1$ by graphing transformations in the indicated stages. Clearly indicate 3 points on each curve by putting in an appropriate dot. **Note:** The first set of axes should have 3 clearly drawn curves, each one labeled. What you are graphing should be put into the 3 blanks following the “ $y =$ ”. The 2nd set of axes should have only your final curve.

Basic curve

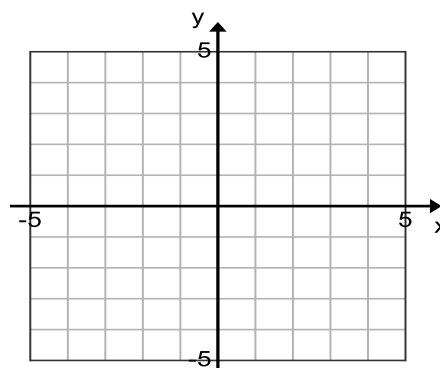
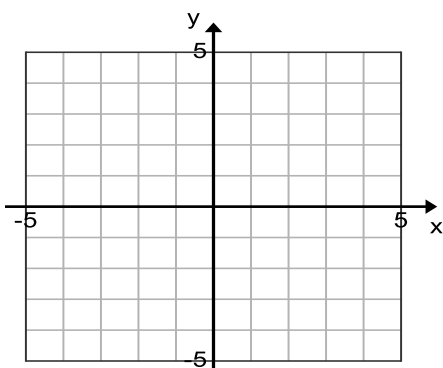
Stretch/Shrink

Reflection: _____

Shift(s): _____

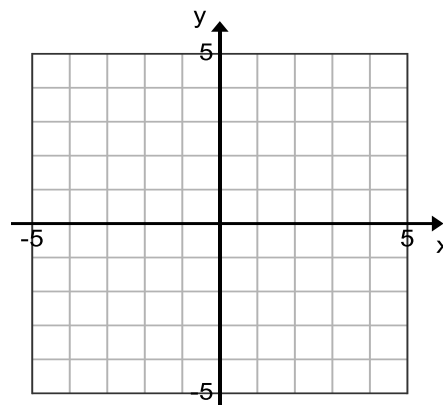
 $y =$ _____ $y =$ _____ $y =$ _____

$$f(x) = -2(x - 3)^2 + 1$$



2. Graph the piecewise-defined function $f(x) = \begin{cases} 5x + 1 & x < 0 \\ 2x^2 - 4 & x \geq 0 \end{cases}$.

State the domain and range



3. Determine the average rate of change of $f(x) = 3x^2 + 3$ on $[-2, 1]$.

4. Given $f(x) = 3x^2 - 4x + 2$, find

a. $f(x + h)$

b. $\frac{f(x+h)-f(x)}{h}$

5. Given $f(x) = \frac{5}{x+3}$ and $g(x) = \frac{2}{x-1}$, find the following and state their domains:

a. $(f - g)(x)$

b. $(f \circ g)(x)$

c. $(\frac{f}{g})(4)$

6. Find the inverse function of $f(x) = \frac{3x-1}{2x+5}$ and determine its range.

7. Use interval notation to write the interval(s) over which $f(x)$ is increasing or decreasing. Identify the value and location of any relative maxima or minima of the function. Assume the graph continues past the grid.

Increasing: _____

Decreasing: _____

Relative minima:

Relative maxima

